Notes Power Model D-MIMO

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I. Central MIMO Power Model

We adopt the parametric power model for upper mid-band (FR3) base stations of [1]. The base station (BS) is equipped with M_{ant} antennas, an equal number of power amplifiers (PAs, used in the downlink) and low-noise amplifiers (LNAs, used in the uplink), and M_{RF} radio-frequency (RF) chains. It serves K single-antenna users over a bandwidth B at carrier frequency f_c using OFDM and space-division multiple access. Setting $M_{\mathrm{RF}} = M_{\mathrm{ant}}$ selects fully-digital beamforming; $M_{\mathrm{RF}} < M_{\mathrm{ant}}$ selects hybrid partially-connected beamforming with $M_{\mathrm{ant}}/M_{\mathrm{RF}}$ phase shifters per RF chain.

A. Model structure

The total consumed power is split into three hardware blocks; digital signal processing, analog signal processing, and the PAs, each divided by its supply-and-cooling efficiency $\eta_{c,\mathrm{sc}} \in (0,1]$:

$$P_{\mathrm{cons}} = \frac{\overline{P_{\mathrm{dig}}}}{\eta_{\mathrm{dig,sc}}} + \frac{\overline{P_{\mathrm{ana}}}}{\eta_{\mathrm{ana,sc}}} + \frac{\overline{P_{\mathrm{PA}}}}{\eta_{\mathrm{PA,sc}}}. \tag{1}$$

Each block is built from the working-mode (active) consumption of its subcomponents, averaged over the operating modes within a frame. We first define this frame averaging (Section I-B) and then reuse it as a function for every component (Sections I-C to I-E).

B. Frame structure and operating-mode averaging

A frame of duration T_f contains a downlink (DL) and an uplink (UL) phase under time division duplexing (TDD), with duration ratios $\tau_{\mathrm{DL}}, \tau_{\mathrm{UL}} \in [0,1]$ and $\tau_{\mathrm{UL}} = 1 - \tau_{\mathrm{DL}}$. Within each direction $i \in \{\mathrm{DL}, \mathrm{UL}\}$ the time is further subdivided into three operating modes, plus an idle mode when the direction is inactive:

- *reference signalling*, for a time fraction $\tau_i\, \tau_{i,\mathrm{sig}}$;
- *data* transmission/reception, for a fraction $\tau_i\,(1 - \tau_{i,\mathrm{sig}})\,\dfrac{N_{a,i}}{N_i}$;
- *micro-sleep* (DTX), for a fraction $\tau_i\,(1-\tau_{i,\mathrm{sig}})\Big(1 - \dfrac{N_{a,i}}{N_i}\Big)$;
- *idle*, for a fraction $(1-\tau_i)$,

where N_i is the number of OFDM symbols (time slots) of direction i and $N_{a,i} \le N_i$ the subset that are *active* (carrying data). The *active-resource time ratio* $N_{a,i}/N_i$ is what sets the data-versus-micro-sleep split: a slot is either active or not, regardless of how many users share it. with $\tau_{i,\mathrm{sig}} \in [0,1]$ the signalling fraction of direction i.

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a) Average physical resource load: The data-mode weight is more conveniently expressed through the *average physical resource load* $\bar{x}_i$, which we now relate to the time ratio $N_{a,i}/N_i$. Let $K_{\max}$ be the maximum number of users that can be served (the number of spatial streams) and $Q_{\max}$ the number of available data subcarriers per active OFDM symbol. The instantaneous load of user k in time slot n is the ratio of utilised to available data subcarriers,

$$\ell_i[k,n] = \frac{Q_i[k,n]}{Q_{\max}}, \qquad i \in \{\mathrm{DL}, \mathrm{UL}\}, \tag{2}$$

for $k = 1, \ldots, K_{\max}$ and $n = 1, \ldots, N_i$. Averaging over all $K_{\max}$ potential users and N_i slots gives the average physical resource load

$$\bar{x}_i \triangleq \bar{\ell}_i = \frac{1}{N_i} \sum_{n=1}^{N_i} \left(\frac{1}{K_{\max}} \sum_{k=1}^{K_{\max}} \frac{Q_i[k,n]}{Q_{\max}} \right). \tag{3}$$

Under space-division multiple access (no OFDMA), an active user in an active slot is allocated *all* data subcarriers, $Q_i[k,n] = Q_{\max}$. Indexing the K_i active users as $\{1, \ldots, K_i\}$ and the $N_{a,i}$ active slots as $\{1, \ldots, N_{a,i}\}$, each active-user/active-slot pair contributes 1 to (3) and every other pair 0; the double sum equals $K_i N_{a,i}$ and

$$\bar{x}_i = \frac{N_{a,i}}{N_i}\,\frac{K_i}{K_{\max}}. \tag{4}$$

b) Equivalence and modelling assumption: The duty-cycle split in the frame structure is governed by the active-resource time ratio $N_{a,i}/N_i$, whereas the power model uses the average load $\bar{x}_i$ as the data-mode weight. Comparing (4) with the data fraction $\tau_i(1-\tau_{i,\mathrm{sig}})\,N_{a,i}/N_i$, the two agree,

$$\bar{x}_i = \frac{N_{a,i}}{N_i} \quad \Longleftrightarrow \quad \frac{K_i}{K_{\max}} = 1 \quad \Longleftrightarrow \quad K_i = K_{\max}, \tag{5}$$

i.e. only when every spatial stream is scheduled. When the maximum number of users is active the average load is simply the ratio of active to total slots; for $K_i < K_{\max}$ the average load (4) is the true active fraction scaled by $K_i/K_{\max} < 1$. The number of active users K_i separately sets the transmit power and the achievable rate, but not whether a slot is active. We use $\bar{x}_i$ as the data-mode weight throughout (and $1-\bar{x}_i$ as the micro-sleep/DTX fraction); this is therefore *exact* in the full-spatial-load regime $K_i = K_{\max}$ evaluated here and an approximation otherwise.

In micro-sleep and idle a component drops to a fraction of its working consumption, set by the reduction factors $\delta_c^{\mu}, \delta_c^{0} \in [0,1]$. For a component c with working power P^{a} during data, P^{s} during signalling, and base power P^{0} scaled in the reduced modes, the frame-averaged consumption of direction i is the time-weighted sum

$$\mathcal{A}_i^c\big(P^{\rm a},P^{\rm s},P^0\big) \triangleq \bar{x}_i\,\tau_i\,(1-\tau_{i,\rm sig})\,P^{\rm a} + \tau_i\,\tau_{i,\rm sig}\,P^{\rm s} + (1-\bar{x}_i)\,\tau_i\,(1-\tau_{i,\rm sig})\,\delta_c^{\mu}\,P^0 + (1-\tau_i)\,\delta_c^{0}\,P^0. \quad (6)$$

For the digital and analog blocks the consumption is identical in all working modes, $P^{\rm a}=P^{\rm s}=P^0=P$, and we use the shorthand $\mathcal{A}_i^c(P)\triangleq\mathcal{A}_i^c(P,P,P)$. Only the PA distinguishes the three levels (Section I-C).

Each direction's hardware is averaged over the *whole* frame, and the two directions are accounted for separately, e.g. for the digital block $\overline{P_{\rm dig}}=\mathcal{A}_{\rm DL}^{\rm dig}(P_{\rm dig,DL})+\mathcal{A}_{\rm UL}^{\rm dig}(P_{\rm dig,UL})$. The idle term $(1-\tau_i)\,\delta_c^0\,P^0$ is therefore the reduced consumption of direction i's chain while that direction is *not* active, i.e. while the opposite direction occupies the frame. For $i=\rm DL$ this fraction equals $\tau_{\rm UL}$ only because the gap-free TDD frame satisfies $\tau_{\rm UL}=1-\tau_{\rm DL}$; it represents the idle DL chain, not the active UL processing, which is captured by the separate $\mathcal{A}_{\rm UL}$ term.

Remark 1 (Load-dependent split)**.** *Only the data term in* (6) *scales with the load $\bar{x}_i$, while the micro-sleep term scales with $(1-\bar{x}_i)$. Collecting the terms proportional to $\bar{x}_i$ separates each averaged consumption into a* load-dependent *part and a* load-independent *part, which is used in the analysis to attribute consumption to the delivered traffic.*

C. Power amplifiers

The PAs transmit only in the DL. With a per-PA output power $P_a=P_{T,\rm DL}/M_{\rm ant}$, where $P_{T,\rm DL}$ is the total DL transmit power, the instantaneous consumption of one PA is modelled as

$$P_{\rm PA}(p)=\frac{1}{\eta_{\rm PA,max}}\left[\xi\,P_{\max}+(1-\xi)\,P_{\max}^{1-\alpha}\,p^{\alpha}\right], \quad (7)$$

where p is the output power, $P_{\max}$ the maximum output power (at a fixed back-off from saturation), $\eta_{\rm PA,max}$ the maximum PA efficiency, $\xi\in[0,1]$ the static-to-dynamic consumption ratio, and $\alpha\in[0,1]$ the efficiency exponent. Averaging over the DL frame and summing over the $M_{\rm ant}$ PAs gives

$$\overline{P_{\rm PA}}=M_{\rm ant}\,\mathcal{A}_{\rm DL}^{\rm PA}\big(P_{\rm PA}(P_a),\,P_{\rm PA}(\zeta_{\rm sig}P_{\max}),\,P_{\rm PA}(0)\big), \quad (8)$$

where $\zeta_{\rm sig}\in(0,1]$ is the fraction of $P_{\max}$ used during signalling, and the zero-output consumption $P_{\rm PA}(0)$ is the base level scaled in micro-sleep and idle.

D. Analog signal processing

The analog block comprises, per RF chain, the DACs (DL) or ADCs (UL), RF filters, mixers, $M_{\rm PS}$ phase shifters, and the LNAs (UL); the local oscillator (LO) is shared across chains. With $M_{\rm PS}=M_{\rm ant}/M_{\rm RF}$ for hybrid beamforming ($M_{\rm PS}=0$ for fully-digital), the working-mode consumption per direction is

$$P_{\rm ana,DL}=P_{\rm LO}+M_{\rm RF}\big(2P_{\rm DAC}+2P_{\rm filt,RF}+2P_{\rm mix}+M_{\rm PS}P_{\rm PS}\big), \quad (9)$$

$$P_{\rm ana,UL}=P_{\rm LO}+M_{\rm RF}\Big(2P_{\rm ADC}+2P_{\rm filt,RF}+2P_{\rm mix}+M_{\rm PS}P_{\rm PS}+\tfrac{M_{\rm ant}}{M_{\rm RF}}P_{\rm LNA}\Big), \quad (10)$$

with the factor 2 accounting for the in-phase and quadrature branches. The subcomponents are modelled as

$$P_{\rm DAC}=\Xi_{\rm DAC,1}2^{b_{\rm DAC}}+\Xi_{\rm DAC,2}\,b_{\rm DAC}f_{\rm DAC}, \quad (11)$$

$$P_{\rm ADC}=W_{\rm ADC}\,f_{\rm ADC}\,2^{b_{\rm ADC}}, \quad (12)$$

$$P_{\rm mix}=\Xi_{\rm mix}f_c, \quad (13)$$

$$P_{\rm PS}=\Xi_{\rm PS}B, \quad (14)$$

$$P_{\rm LNA}=\Xi_{\rm LNA}B, \quad (15)$$

where $b_{\rm DAC},b_{\rm ADC}$ are the effective resolutions, $f_{\rm DAC},f_{\rm ADC}$ the converter sampling rates, $W_{\rm ADC}$ the ADC Walden figure-of-merit, and $\Xi_\bullet$ the respective figures-of-merit. The frame-averaged analog consumption is

$$\overline{P_{\rm ana}}=\mathcal{A}_{\rm DL}^{\rm ana}(P_{\rm ana,DL})+\mathcal{A}_{\rm UL}^{\rm ana}(P_{\rm ana,UL})+P_{\rm ana,misc}. \quad (16)$$

E. Digital signal processing

In the DL the digital block runs the channel encoder, MIMO precoder, IFFT, DPD, and baseband (BB) filter; in the UL the channel decoder, MIMO combiner, FFT, and BB filter. Each subcomponent consumption is a static (signal-independent) term plus a dynamic term equal to its complex giga-operations-per-second (GOPS) Ξ divided by the computational efficiency η (in GOPS/W) of the chosen FPGA/ASIC implementation:

$$P_{\rm enc}=P_{\rm enc,s}+\frac{f_{s,\rm I}}{\eta_{\rm enc}}\frac{14}{24}\frac{R_{\rm DL}}{\tilde{B}}, \quad (17)$$

$$P_{\rm dec}=P_{\rm dec,s}+\frac{f_{s,\rm I}}{\eta_{\rm dec}}\frac{175}{6}\frac{R_{\rm UL}}{\tilde{B}}, \quad (18)$$

$$P_{\rm pre}=\bar{M}_{\rm RF}P_s+\frac{f_{s,\rm I}}{\eta_{\rm pre}}M_{\rm RF}\Xi_{\rm pre}, \quad (19)$$

$$P_{\rm comb}=\bar{M}_{\rm RF}P_s+\frac{f_{s,\rm I}}{\eta_{\rm comb}}M_{\rm RF}\,2K, \quad (20)$$

$$P_{\rm IFFT}=P_{\rm IFFT,s}+\frac{f_{s,\rm II}}{\eta_{\rm IFFT}}M_{\rm RF}\tfrac{3}{2}\log_2 Q_{\rm IFFT}, \quad (21)$$

$$P_{\rm DPD}=P_{\rm DPD,s}+\frac{f_{s,\rm II}}{\eta_{\rm DPD}}M_{\rm RF}\Xi_{\rm DPD}, \quad (22)$$

$$P_{\rm filt,BB}=\bar{M}_{\rm RF}P_s+\frac{f_{s,\rm II}}{\eta_{\rm BB}}M_{\rm RF}\Xi_{\rm BB}, \quad (23)$$

where $R_{\rm DL},R_{\rm UL}$ are the delivered DL/UL sum rates, $\tilde{B}$ the effective bandwidth, $\bar{M}_{\rm RF}=\lceil M_{\rm RF}/32\rceil$ the number of FPGAs (one per 32 chains, sharing the precoder/combiner/BB-filter static power P_s), and $Q_{\rm IFFT}$ the IFFT/FFT size. The two sampling rates are the constellation rate $f_{s,\rm I}=\mu B$ (for encoding/decoding and MIMO processing) and the oversampled rate $f_{s,\rm II}=Q_{\rm IFFT}\Delta f$ (for the OFDM, DPD, and BB operations), with $\mu\in[0,1]$ and subcarrier spacing Δf. For the zero-forcing

MIMO processor exploiting channel reciprocity, the precoder GOPS is

$$\Xi_{\mathrm{pre}} = 2K + \frac{1}{\upsilon_{\mathrm{coh}}}\left(\frac{K^3}{3M_{\mathrm{RF}}} + 3K^2 + K\right), \tag{24}$$

with υ_{coh} the number of samples per channel coherence block over which the precoder is reused; the combiner only applies the precoding matrix, giving $\Xi_{\mathrm{comb}} = 2K$. The active digital consumption per direction is the sum of its subcomponents,

$$P_{\mathrm{dig,DL}} = P_{\mathrm{enc}} + P_{\mathrm{pre}} + P_{\mathrm{IFFT}} + P_{\mathrm{DPD}} + P_{\mathrm{filt,BB}}, \tag{25}$$

$$P_{\mathrm{dig,UL}} = P_{\mathrm{dec}} + P_{\mathrm{comb}} + P_{\mathrm{IFFT}} + P_{\mathrm{filt,BB}}, \tag{26}$$

and the frame-averaged digital consumption is

$$\overline{P_{\mathrm{dig}}} = \mathcal{A}^{\mathrm{dig}}_{\mathrm{DL}}(P_{\mathrm{dig,DL}}) + \mathcal{A}^{\mathrm{dig}}_{\mathrm{UL}}(P_{\mathrm{dig,UL}}). \tag{27}$$

F. Delivered rate and energy efficiency

The achievable ergodic sum rate of direction i, which drives the encoder/decoder consumption in (17), follows from the per-user SINRs under zero-forcing beamforming as

$$R_i = \bar{x}_i\,\tau_i\,(1-\tau_{i,\mathrm{sig}})\,\frac{\tilde{B}}{q_i}\,\mathbb{E}\left\{\sum_{\nu=1}^{q_i}\sum_{k=1}^{K}\log_2(1+\mathrm{SINR}_{i,k}[\nu])\right\}, \tag{28}$$

where the pre-log collects the load, duration, signalling-overhead, and effective-bandwidth-per-subcarrier factors. The base-station energy efficiency then follows as the delivered rate per consumed watt,

$$\mathrm{EE} = \frac{R_{\mathrm{DL}} + R_{\mathrm{UL}}}{P_{\mathrm{cons}}} \quad [\mathrm{bit/J}]. \tag{29}$$

II. Distributed Massive MIMO System Model

A. Downlink System Model

We consider the downlink of a distributed (cell-free) massive multiple input multiple output (MIMO) network in which L access points (APs), each equipped with M antennas, jointly serve K single-antenna users over the same time-frequency resources. The APs are connected through fronthaul links to a central processing unit (CPU) that encodes the data and computes the precoders, so that the LM antennas act as a single distributed array. Transmission uses OFDM with Q data subcarriers, indexed by $q = 1, \ldots, Q$; the model below is stated per subcarrier and the same processing is applied on each of them.

Let $\mathbf{h}_{kl}[q] \in \mathbb{C}^M$ denote the channel from AP l to user k on subcarrier q. Stacking the contributions of all APs gives the collective channel

$$\mathbf{h}_k[q] = \begin{bmatrix} \mathbf{h}_{k1}[q] \\ \vdots \\ \mathbf{h}_{kL}[q] \end{bmatrix} \in \mathbb{C}^{LM}. \tag{30}$$

The channel realizations are drawn from the 3GPP TR 38.901 urban-microcell (UMi) geometry-based stochastic channel model [2], using the implementation provided by the Sionna link-level simulator [3]. Each AP is modelled as a 38.901 base station carrying an M-element half-wavelength uniform linear array (ULA) of omnidirectional elements and each user as a single-antenna terminal. Every AP-user link is assigned a line-of-sight or non-line-of-sight condition according to the distance-dependent LOS probability of the UMi scenario, and its distance-dependent path loss, log-normal shadow fading, and small-scale multipath then determine $\mathbf{h}_{kl}[q]$.

Because the model is specified directly in terms of channel realizations rather than through a closed-form spatial correlation matrix, the large-scale fading coefficient β_{kl} between AP l and user k, which drives the downlink power control of Section II-B, is obtained from the realizations as the average per-antenna channel gain,

$$\beta_{kl} = \frac{1}{M}\,\mathbb{E}\left(\|\mathbf{h}_{kl}[q]\|^2\right), \tag{31}$$

which is estimated by averaging $|[\mathbf{h}_{kl}[q]]_m|^2$ over the M antennas of AP l and the Q subcarriers. It captures the combined effect of path loss and shadow fading (and, on a LOS link, of the specular component) and so summarizes the slow, position-dependent part of the channel that the power allocation acts on.

The CPU maps the data symbol $s_k[q] \in \mathbb{C}$ of user k, with unit power $\mathbb{E}\left(|s_k[q]|^2\right) = 1$ and mutually independent across users, onto the M antennas of each AP. AP l transmits

$$\mathbf{x}_l[q] = \sum_{k=1}^{K} \mathbf{w}_{kl}[q]\,s_k[q] = \mathbf{W}_l[q]\,\mathbf{s}[q], \tag{32}$$

where $\mathbf{w}_{kl}[q] \in \mathbb{C}^M$ is the precoding vector that AP l applies to user k, $\mathbf{s}[q] = [\,s_1[q]\ \cdots\ s_K[q]\,]^\mathsf{T} \in \mathbb{C}^K$ collects the data symbols, and $\mathbf{W}_l[q] = [\,\mathbf{w}_{1l}[q]\ \cdots\ \mathbf{w}_{Kl}[q]\,] \in \mathbb{C}^{M\times K}$ is the precoding matrix of AP l. Collecting the per-AP precoders of a given user into $\mathbf{w}_k[q] = [\,\mathbf{w}_{k1}[q]^\mathsf{T}\ \cdots\ \mathbf{w}_{kL}[q]^\mathsf{T}\,]^\mathsf{T} \in \mathbb{C}^{LM}$ yields the network-wide precoding matrix

$$\mathbf{W}[q] = \left[\,\mathbf{w}_1[q]\ \cdots\ \mathbf{w}_K[q]\,\right] \in \mathbb{C}^{LM\times K}, \tag{33}$$

whose l-th block of M rows equals $\mathbf{W}_l[q]$, so that the stacked transmit vector is $\mathbf{x}[q] = [\,\mathbf{x}_1[q]^\mathsf{T}\ \cdots\ \mathbf{x}_L[q]^\mathsf{T}\,]^\mathsf{T} = \mathbf{W}[q]\,\mathbf{s}[q] \in \mathbb{C}^{LM}$.

The signal received by user k on subcarrier q is $y_k[q] = \mathbf{h}_k[q]^H\mathbf{x}[q] + n_k[q]$, which, using (33), separates into the desired term, the multi-user interference, and the noise,

$$y_k[q] = \underbrace{\mathbf{h}_k[q]^H\mathbf{w}_k[q]\,s_k[q]}_{\text{desired signal}} + \underbrace{\sum_{\substack{i=1\\ i\neq k}}^{K}\mathbf{h}_k[q]^H\mathbf{w}_i[q]\,s_i[q]}_{\text{multi-user interference}} + \underbrace{n_k[q]}_{\text{noise}}, \tag{34}$$

where $n_k[q] \sim \mathcal{CN}\left(0, \sigma^2\right)$ is the receiver noise with power σ^2, independent across users and subcarriers and independent of the channels and data. The scalar $\mathbf{h}_k[q]^H\mathbf{w}_k[q]$ is the effective channel seen by user k: the distributed precoder turns the LM-antenna channel into an equivalent single-antenna link.

Treating the multi-user interference in (34) as noise, the effective downlink signal-to-interference-plus-noise ratio (SINR) of user k on subcarrier q is

$$\mathrm{SINR}_k^{\mathrm{dl}}[q] = \frac{\left|\mathbf{h}_k[q]^H \mathbf{w}_k[q]\right|^2}{\sum_{\substack{i=1 \\ i\neq k}}^{K} \left|\mathbf{h}_k[q]^H \mathbf{w}_i[q]\right|^2 + \sigma^2}. \tag{35}$$

With the effective channel known at the receiver, an achievable rate for user k on subcarrier q is $\log_2\big(1+\mathrm{SINR}_k^{\mathrm{dl}}[q]\big)$, and the delivered downlink sum rate R_{DL} follows by inserting these per-user, per-subcarrier SINRs into the ergodic rate expression of Section I-F ((28)), which in turn drives the encoder and MIMO processing consumption of the power model.

B. Precoder and Power Control Designs

Unlike co-located massive MIMO, where a single sum-power budget applies to the whole array, each AP is a separate transmitter and is subject to its own power constraint. Since the data symbols are independent and unit-power, the average power radiated by AP l across the OFDM block is

$$\sum_{q=1}^{Q} \mathbb{E}\left(\|\mathbf{x}_l[q]\|^2\right) = \sum_{q=1}^{Q}\sum_{k=1}^{K} \mathbb{E}\left(\|\mathbf{w}_{kl}[q]\|^2\right) \le \rho_{\max}, \tag{36}$$

$$l = 1, \dots, L, \tag{37}$$

where $\rho_{\max}$ is the maximum downlink transmit power per AP, assumed equal for all APs, and the expectation is taken over the data and channel realizations.

It is convenient to split each precoder into a transmit direction and a power control coefficient,

$$\mathbf{w}_k[q] = \sqrt{\rho_k}\, \frac{\bar{\mathbf{w}}_k[q]}{\sqrt{\mathbb{E}\left(\|\bar{\mathbf{w}}_k[q]\|^2\right)}}, \tag{38}$$

where $\bar{\mathbf{w}}_k[q] \in \mathbb{C}^{LM}$ is an arbitrarily scaled vector that fixes the spatial direction of the transmission to user k and $\rho_k \ge 0$ is the downlink power allocated to that user, so that $\mathbb{E}\left(\|\mathbf{w}_k[q]\|^2\right) = \rho_k$. This separation isolates two design tasks, the choice of the precoding directions $\{\bar{\mathbf{w}}_k[q]\}$ and the power allocation that ties them to the per-AP budget in (36). How both are computed depends on the level of cooperation between the APs, so the two operation modes are treated in turn: centralized operation (Section II-B1), where the CPU designs all precoders jointly and assigns one power coefficient per user, and local operation (Section II-B2), where each AP designs its own precoder and sets its powers from local information alone. In both cases the allocation depends mainly on the large-scale fading of (31) and can be held fixed over many coherence blocks, whereas the directions are recomputed every block.

1) Centralized Operation: In centralized operation the APs forward their received uplink pilots to the CPU, which estimates the collective channels $\{\mathbf{h}_k[q]\}$ and designs all precoders jointly [4]. Exploiting TDD reciprocity, these uplink estimates serve as the downlink channel state information (CSI), so no downlink pilots are needed. Because the CPU sees the whole LM-antenna array, it can make the serving APs cancel one another's interference, using up to LM spatial degrees of freedom per user. Collecting the user channels into $\mathbf{H}[q] = [\,\mathbf{h}_1[q] \;\cdots\; \mathbf{h}_K[q]\,] \in \mathbb{C}^{LM\times K}$, the zero forcing (ZF) precoder picks the directions that null all inter-user interference,

$$\bar{\mathbf{W}}[q] = \mathbf{H}[q]\big(\mathbf{H}[q]^H\mathbf{H}[q]\big)^{-1} \in \mathbb{C}^{LM\times K}, \tag{39}$$

whose column $\bar{\mathbf{w}}_k[q]$ is orthogonal to every other user's channel when the CSI is accurate; this requires $K \le LM$ so that the Gram matrix $\mathbf{H}[q]^H\mathbf{H}[q]$ is invertible. Zero-forcing disregards the noise, so at low SNR it is improved by adding a diagonal loading λ to the Gram matrix, giving the regularized family

$$\bar{\mathbf{W}}[q] = \mathbf{H}[q]\big(\mathbf{H}[q]^H\mathbf{H}[q] + \lambda\mathbf{I}_K\big)^{-1}, \tag{40}$$

which trades a little residual interference for robustness. With the loading set to the noise power, $\lambda = \sigma^2$, this is the downlink minimum mean squared error (MMSE) precoder, the dual of MMSE combining; under the perfect-CSI assumption used here the MMSE precoder coincides with regularized zero forcing (RZF) at that loading, and the two differ only once a channel-estimation-error covariance is modelled. Zero-forcing, the $\lambda = 0$ limit, is the scheme adopted in the power model of Section I.

Because a centralized precoder is designed jointly across the APs, a single power coefficient $\rho_k \ge 0$ is attached to each user and is common to all APs; the direction $\bar{\mathbf{w}}_k[q]$ already fixes how that power is split over the LM antennas, and the direction is scaled to radiate ρ_k as in (38). Two nominal allocations set $\{\rho_k\}$, both spending the total network budget $P_{\mathrm{tot}} = L\rho_{\max}$. Equal power gives every user the same share, $\rho_k = P_{\mathrm{tot}}/K$, whereas fractional power control weights users by their aggregate large-scale gain $\beta_k = \sum_{l=1}^{L}\beta_{kl}$,

$$\rho_k = P_{\mathrm{tot}}\, \frac{\beta_k^{v}}{\sum_{i=1}^{K}\beta_i^{v}}, \tag{41}$$

with the exponent v trading throughput ($v > 0$, favouring strong users) against fairness ($v < 0$, protecting weak users). These nominal powers make the per-AP loads average to $\rho_{\max}$ but do not by themselves respect each per-AP constraint in (36). Since the joint directions cannot be rescaled per AP without breaking the interference nulling, a single common factor

$$\gamma = \sqrt{\frac{\rho_{\max}}{\max_l \tilde{p}_l}}, \qquad \tilde{p}_l = \sum_{q=1}^{Q}\sum_{k=1}^{K}\mathbb{E}\left(\|\mathbf{w}_{kl}[q]\|^2\right), \tag{42}$$

rescales all precoders, where $\tilde{p}_l$ is the power AP l would radiate under the nominal allocation. The busiest AP then transmits at exactly $\rho_{\max}$ and the others stay below their budget, while the common scalar preserves the relative amplitudes and phases across APs and hence the interference nulling of (39). Because $\sum_k \rho_k = L\rho_{\max}$, the per-AP powers average to $\rho_{\max}$, so $\gamma \leq 1$ and the constraint is always feasible.

2) Local Operation: In local (distributed) operation the CPU only encodes the data, and each AP forms its own precoding directions from its locally estimated channels $\hat{\mathbf{h}}_{kl}[q]$, without sharing instantaneous CSI. An AP can then suppress only the interference it itself generates, using the M spatial degrees of freedom of its own array, so the coherent cross-AP cancellation of the centralized case is lost. The simplest choice is maximum-ratio transmission (maximum ratio (MR)), which points each beam along the local channel,

$$\bar{\mathbf{w}}_{kl}[q] = \mathbf{h}_{kl}[q], \tag{43}$$

maximizing the signal power delivered to the desired user but ignoring the interference it causes. Interference is instead suppressed locally by the counterpart of zero-forcing applied to each AP's own users. Writing $\mathbf{E}_l[q] = [\,\mathbf{h}_{1l}[q] \ \cdots \ \mathbf{h}_{Kl}[q]\,] \in \mathbb{C}^{M\times K}$ for AP l's local channel matrix, the per-AP regularized precoder is

$$\mathbf{W}_l[q] = \left(\mathbf{E}_l[q]\mathbf{E}_l[q]^H + \lambda \mathbf{I}_M\right)^{-1}\mathbf{E}_l[q] \in \mathbb{C}^{M\times K}, \tag{44}$$
$$l = 1, \ldots, L, \tag{45}$$

whose column k is the local direction $\bar{\mathbf{w}}_{kl}[q]$, and the collective direction $\bar{\mathbf{w}}_k[q]$ stacks the L per-AP blocks. Local zero-forcing is the $\lambda = 0$ limit, which nulls the interference an AP causes to its own users only when $M \geq K$; because each AP is small (typically $M < K$), the loading is needed to keep the $M \times M$ inverse well conditioned, giving local RZF for a heuristic λ and local MMSE for $\lambda = \sigma^2$. As in the centralized case, under perfect CSI local MMSE coincides with local RZF at that loading. The $M\times M$ antenna-domain form is used because each AP's array is small.

Because each AP designs and normalizes its own block independently, the power is parametrized per user per AP: AP l normalizes its power to respect $\rho_{kl} \geq 0$ for user k this is achieved as

$$\mathbf{w}_{kl}[q] = \sqrt{\rho_{kl}}\,\bar{\mathbf{w}}_{kl}[q] / \sqrt{\mathbb{E}\left(\|\bar{\mathbf{w}}_{kl}[q]\|^2\right)}. \tag{46}$$

Fractional power control is used, each AP splits its own budget $\rho_{\max}$ among its users according to their local large-scale gains,

$$\rho_{kl} = \rho_{\max} \frac{\beta_{kl}^v}{\sum_{i=1}^K \beta_{il}^v}, \qquad l = 1, \ldots, L, \tag{47}$$

so that $\sum_{k=1}^K \rho_{kl} = \rho_{\max}$ by construction and every AP meets its per-array budget in (36) with equality, using only local quantities and without any CPU-side global rescaling. The exponent v plays the same throughput-versus-fairness role as above, with $v = 1/2$ recovering the heuristic of [4]. Beyond such statistics-based rules the powers could be optimized for a network utility such as max-min fairness or sum rate, but those problems do not scale to very large networks, so the heuristics above are the practical default.

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